

CHALLENGE – 14 – CRYPTOGRAPHY

basic-mod1 

Medium Cryptography picoCTF 2022

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Description

We found this weird message being passed around on the servers, we think we have a working decryption scheme.

Download the message [here](#).

Take each number mod 37 and map it to the following character set: 0-25 is the alphabet (uppercase), 26-35 are the decimal digits, and 36 is an underscore.

Wrap your decrypted message in the picoCTF flag format (i.e. `picoCTF{decrypted_message}`).

Hints

1 2

`mod 37` means modulo 37. It gives the remainder of a number after being divided by 37.

24,848 users solved



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 picoCTF{FLAG}Submit
FlagSolution
Walkthrough1. Understand the
Decryption
Scheme

The challenge provides a specific algorithm for decryption:

1. Take each number in the provided list and calculate its value **modulo 37** ($x \pmod{37}$).

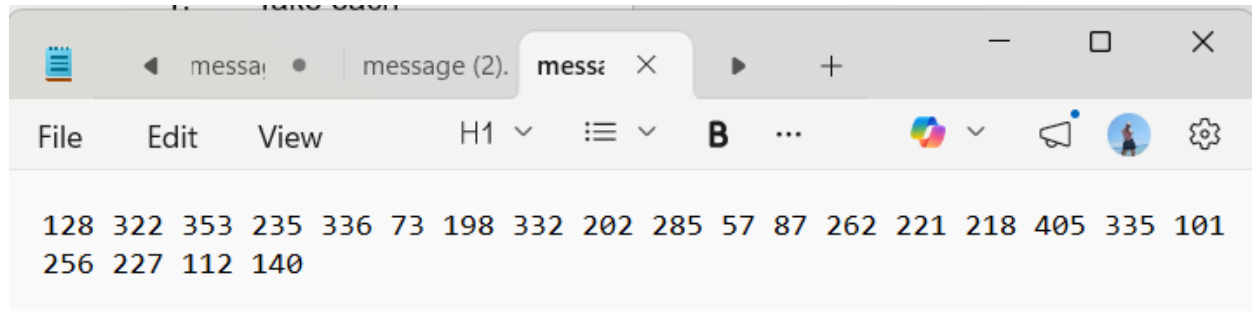
2. Map the resulting remainder to a character based on these

rules:

- **0–25:** Uppercase Alphabet (A–Z, where 0 = A).
- **26–35:** Decimal Digits (0–9).
- **36:** Underscore (_).

2. Extract Data and Perform Calculations

The provided message contains the following numerical sequence:



A Python script was used to automate the modulo operations:

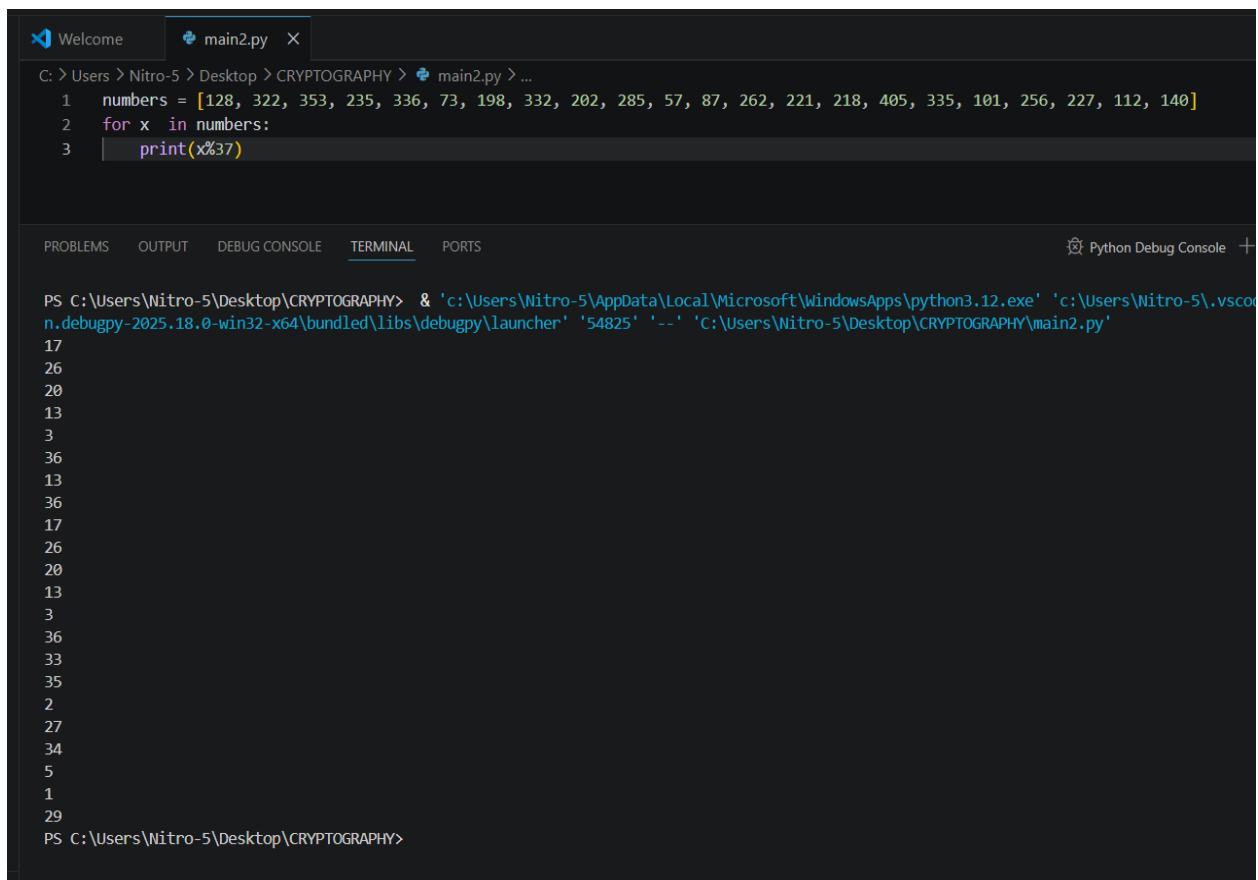
Python

```
numbers = [128, 322, 353, 235, 336, 73, 198, 332, 202, 285, 57, 87, 262, 221, 218, 405, 335, 101, 256, 227, 112, 140]
```

```
for x in numbers:
```

```
    print(x % 37)
```

The resulting remainders are: 17, 26, 20, 13, 3, 36, 13, 36, 17, 26, 20, 13, 3, 36, 33, 35, 2, 27, 34, 5, 1, 29



3. Map to Character Set

By applying the mapping rules to the remainders, the plaintext is revealed:

- 17 -> R
- 26 -> 0 (The first digit in the 26–35 range)
- 20 -> U
- 13 -> N
- 3 -> D
- 36 \rightarrow \text{ }
- ...and so on.

The decrypted string is:

17
26
20
13
3
36
13
36
17
26
20
13
3
36
33
35
2
27
34
5
1

29 follow this conversion rule map it to the following character set: 0-25 is the alphabet (uppercase), 26-35 are the decimal digits, and 36 is an underscore.



ROUND_N_ROUND_79C18FB3.

4. Final Flag

Wrapping the result in the required picoCTF{} format:
picoCTF{ROUND_N_ROUND_79C18FB3}

- 0-25 → A-Z (0 = A)
- 26-35 → 0-9
- 36 → _

Correct result:

R0UND_N_R0UND_79C18FB3



Reflection

The "basic-mod1" challenge serves as a fundamental introduction to **modular arithmetic** in cryptography. It demonstrates how data can be obscured through mathematical transformations and how a "character map" acts as a bridge between numerical output and human-readable text.

A key takeaway from this exercise is the utility of **automation**. While the modulo operations and character mapping could be done manually, writing a simple script ensures accuracy and speed, which are vital skills in CTF competitions. This challenge highlights that even "medium" difficulty tasks often rely on the precise application of simple mathematical foundations like the modulo operator.